

Tekion FTE Interview Experience

Intro:

-> Tekion is a software development company that develops business applications for automobile industries and dealers.

-> For our batch(2019 - 2023) Placements started by 2nd - 3rd week of August.

-> Tekion came as the 15th company for our Placements and I had one Online Round and 3 technical rounds and one HM discussion and finally HR round.

Online Round: - hackerearth platform

We had 5 coding questions.

1) [Auto Suggest](#) - (Edit Distance ,Levenshtein Distance) - 50 marks

2) URL checking - 20 marks

3) How many patterns - 20 marks

You are given a tree with N nodes. Only one of the nodes in the tree is painted black and the rest are painted white. You can paint any white node as black if it has an edge connecting a black-painted node.

Determine the count of distinct painted trees we can obtain by painting some of the white nodes as black following the condition. Two trees are considered painted differently if there is at least one node painted black in one tree and white in the other.

4) Triplets Count with modification - 50 marks

5) A Remainder Problem - 100 marks

You are given a large number N in the form of a string. An integer K and Q queries are also given.

For each query, you are required to determine whether such subsequences of N exist and following the provided condition:

When the number formed by concatenating the digits of N without changing relative ordering is divided by K then it leaves the remainder X(each number in the queries array).

If you can choose a subsequence then the answer for that query(X) is "Yes" .

Else the answer is "No" for that query(X).

Example:

Input : N = 25 ; K = 5 ; Q = 2 ; queries = {0,1}

Output : ["yes", "no"]

There is a subsequence "5" when divided by 5 gives remainder as 0(X). So, "Yes" for the 1st query.

There is no any subsequence when divided by 5 gives remainder as 2(X). So, "No" for the 2nd query.

For this brute force solution i.e generating all subsequences and checking won't work since the maximum length of the given string 'N' is very very large.

Technical Interview Round - 1: Face-to-Face @CUIC

35 students were shortlisted for the interview Round - 1.

- > This round was more like a discussion.
- > Firstly, the interviewer asked me to write a code for merge sort.
- > Then he asked me to reverse the given number without converting it to a string or an array. ex) $n = 34561$ $rev(n) = 16543$.
- > Then he asked a problem in array, the problem statement goes like this,
Given an array of even length. We can add any number of pairs (n,n) at any place in the array(but two n 's should be added continuously) such that the array must be splitted into two equal parts(0 to $n/2$ and $n/2$ to n). We must find the minimum number of such pairs, if it is not possible to do so then, return -1.
eg) $arr = [1, 2, 2, 1]$
Here we can add $(1,1)$ at the 3rd position so that arr we become like,
 $[1, 2, 1, 1, 2, 1] \Rightarrow$ now first part $(1\ 2\ 1)$ and second part $(1\ 2\ 1)$ of the array is equal, so our answer is 2
eg) $arr = [3, 4, 2, 1]$ here it is not possible, so answer = -1.
- > Then he asked me about the [balanced parenthesis problem](#).
- > Then he asked me about indexing in DBMS and asked how it is implemented? I explained about indexing and B+ trees.
- > Then he asked what I did in my internship?

Technical Interview Round - 2:

20 students were shortlisted for the interview Round - 2.

- > This round was fully based on my projects.
- > I had put one DBMS and one ML and one Intern project in my resume.
- > I explained those 3 projects.
- > Then the interviewer asked what could I add more in the those projects and how could it be implemented.
- > Then he asked many questions from backend and Database since he is from the backend team.
 - > How to overcome sql injection attacks?
 - > How is the user authenticated and where are the authenticated details stored?
 - > How I stored the passwords in my Databases.
 - > Then many follow up questions in backend and Databases.

Technical Interview Round - 3:(only few people had this round)

- > This round was very easy compared to the previous 2 rounds.
- > This round was a DSA + technical subjects discussion round.
- > He asked me to reverse a linked list using recursion and write pseudo code.

- > He asked what DFS and BFS were and asked me to perform inorder, preorder and postorder traversals.
- > He asked me about the Round-Robin Scheduling algorithm.
- > He asked about different keys present in the Database tables.
- > Then He asked whether I had some questions for him.

HM Discussion:

- > This round was a discussion about what we will be doing in tekion.
- > Two people jointly attended this discussion.
- > If we had any doubts or questions about our role, we could ask them to the interviewer(he was the head of the backend team).
- > This round was short and went for around 15 minutes.

HR Round:

- > The interviewer asked me to introduce myself.
- > Then she asked where my native place was located and then she asked about my family background.
- > Then she asked about my strengths and weaknesses and why I call them my strengths.
- > Then there was a one behaviour question like "You have an offer from 2 companies, one is from tekion and another from another company, Apart from money what factors will you consider?"
- > Then she asked whether I had any questions for her.

Learnings and Takeaways:

- > **Keep your expectations to zero.**
- > Stick to your strengths and reduce your weaknesses.
 - > **Minimise your negatives and mistakes** done in your previous interviews and online rounds and try to solve those questions for which you did not find the solution.
- > Have good knowledge in DSA, DBMS, OS, OOPS.
- > Before Any interview, revise all sorting algorithms(imp : **merge sort**), OOPS concepts, SQL queries, DBMS interview questions, OS interview questions.
- > Ask out for help from your seniors and friends when you get stuck(not during the test :)).
 - > Interviewer will also help you when you get stuck.
 - > Prepare answers for some standard HR questions and prepare some questions to ask them to the interviewer.

Finally 12 people were placed in tekion including myself.

All the Very best!

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